

Electrical Characterization of Organic FETs

Temperature analysis with Cryostat

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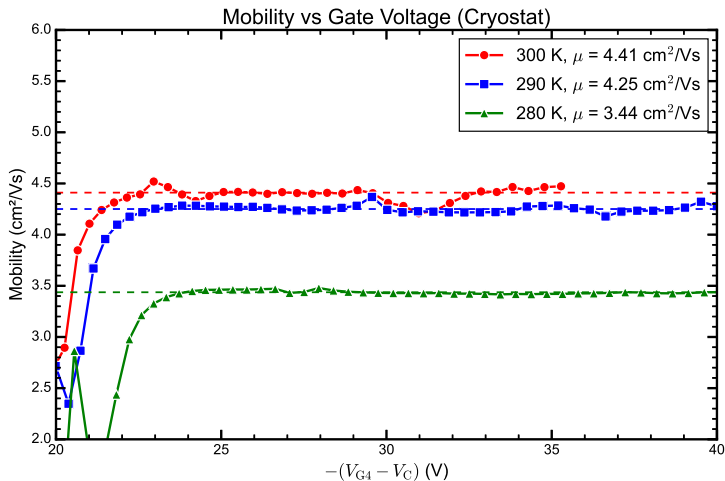


Figure: vdP for different temperatures (PS_PaN 2).

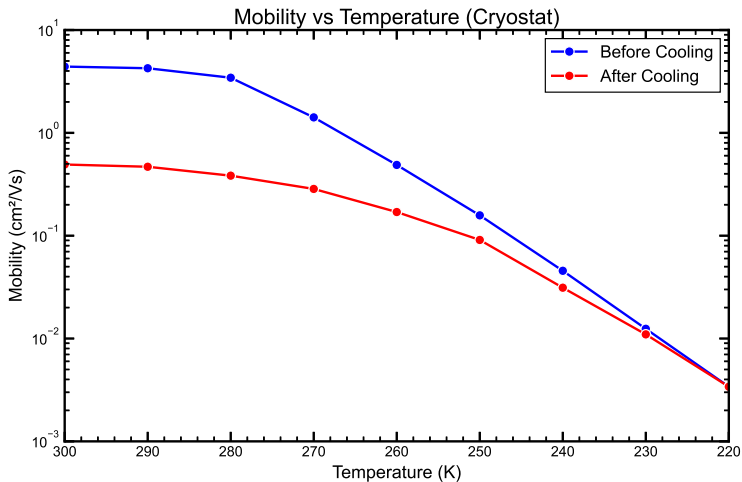


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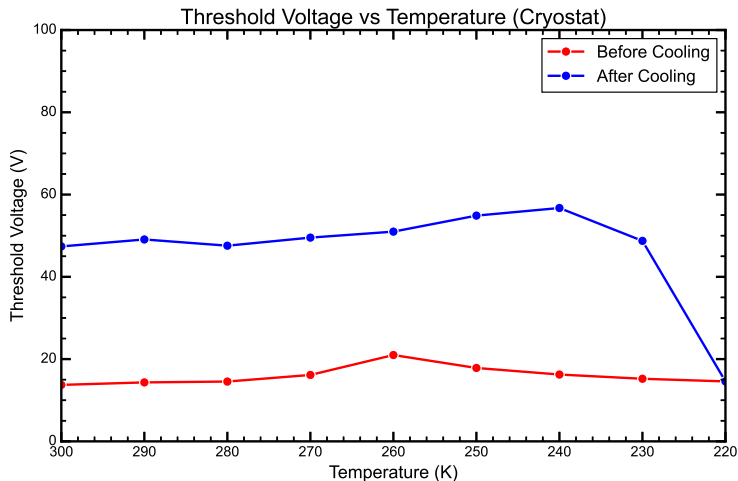


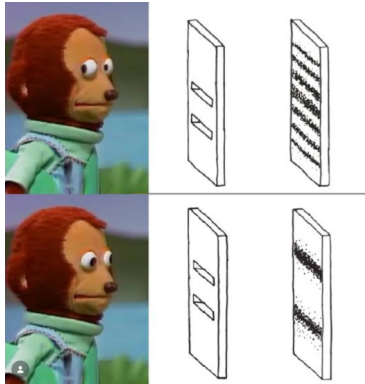
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Observation

- Mobility drops exponentially from 300 K to 220 K.
- Mobility stays lower after cooling (bad hysteresis).
- Threshold voltage shifts upward (also bad hysteresis).

Interpretation

- Cooling creates a *temperature memory* state.
- Likely causes: charge trapping, frozen dipoles, morphology/contact changes, or adsorbates.



Thank you for your attention!
Any questions?